Optimal pressure sensor locations in smart insoles for heel-strike and toe-off detection

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***Abstract*—Heel-strike and toe-off are important gait events for gait analysis. Smart insoles with pressure sensors in different locations have been proposed to detect these two gait events. However, there is no research to reveal optimal sensor locations under the foot for heel-strike and toe-off detection. To address this problem, a novel method based on the mechanism of heel-strike and toe-off detection was proposed to find out the optimal sensor locations. A smart insole system, which has 96 pressure sensors uniformly distributed on it, was used to supply ground truth. Experiment data was collected from six subjects, each of whom wore a pair of smart insoles and walked 15 steps at their comfort speeds. The results showed that two top optimal pressure locations identified with the proposed method were enough to detect heel-strike and toe-off with good accuracy. Options of top optimal sensor locations for heel-strike and toe-off detection were mainly concentrated in the middle of the heel area (sensors 5, 6, 8, 10, 11, 13, and 14) and the front edge of the metatarsal bone area (sensors 55, 59, 60, and 61), respectively. This research provided references for future researchers to effectively use pressure sensors for smart insole design and gait analysis.**

***Index Terms*—pressure sensor, smart insole, gait analysis, sensor location, heel strike**

I. Introduction

Heel-strike and toe-off are important gait events that are frequently used in gait analysis. Heel-strike is the instant when the heel of the leading foot touches the ground. And toe-off is the instant when the toe leaves the ground [1]. With the detected heel-strike and toe-off, important gait parameters including gait cycle time, stance time, swing time, single support time, double support time, etc. can be calculated [2]. In addition, gait symmetry, gait stability, and gait variability can be calculated with gait cycle time, stance time, single support time, etc. [2], [3]. These gait parameters are popularly used in disease diagnosis. For example, for neural system diseases, such as Parkinson's disease (PD), degrees of gait variability is increased with the disease severity [4].

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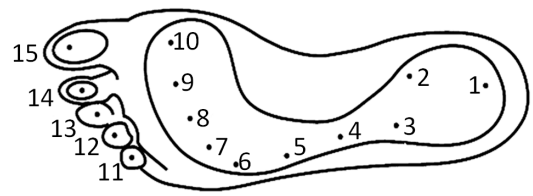

Fig. 1. Foot anatomical area [5]

Since pressure sensors can directly measure the pressure changes during heel-strike and toe-off, they are frequently used for detecting these gait events and doing gait analysis. To make it easy to be applied in real applications, pressure sensors are usually used for the design of wearable systems such as smart insoles. Since the pressure distribution under the foot is not uniform, pressure sensor locations are critical for the performance. Most of the researchers preferred to place pressure sensors on the location with high plantar pressure. According to the foot anatomical area shown in Fig. 1, the foot sole can be divided into 15 areas that support most of the body weight [5]. Those 15 areas shown in Fig. 1 were frequently used as locations to place pressure sensors. Wang et al. proposed a smart insole called FreeWalker which used 8 pressure sensors around areas 1, 2, 3, 5, 7, 9, 10, and 14 to detect heel-strike and toe-off [6]. Sazonov et al. developed a smart insole with 5 pressure sensors around areas 1, 6, 8, 10, and 15 to detect heel-strike, toe-off, stance phase, gait cycles, etc. for recognizing daily living activities such as sitting, standing, walking, running, stair ascent/descent, and cycling [7]. Pappas et al. used 3 pressure sensors around areas 1, 8, and 10 to detect heel-strike, stance, toe-off, and swing for the control of programmable functional electrical stimulation (FES) [8]. Bamberg et al. developed a "GaitShoe" with 4

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pressure sensors around areas 1, 2, 6, and 10 for the detection of heel-strike and toe-off [9].

Although different smart insoles have been developed with pressure sensors for heel-strike and toe-off detection, different research used different numbers of pressure sensors on different locations. For wearable systems, high performance, low cost, and long battery life are important design requirements. To balance these design requirements in the smart insole design, it is important to understand the optimal pressure sensor locations. However, to the best of our knowledge, there is no research that reveals optimal sensor locations for detecting heel-strike and toe-off. To address this problem, a novel method based on the mechanism of heel-strike and toe-off detection was proposed in this research to find out the optimal sensor locations.

II. Methods

In this section, the smart insole system and the method used to identify the optimal pressure sensor locations were specified.

A. Smart Insole System

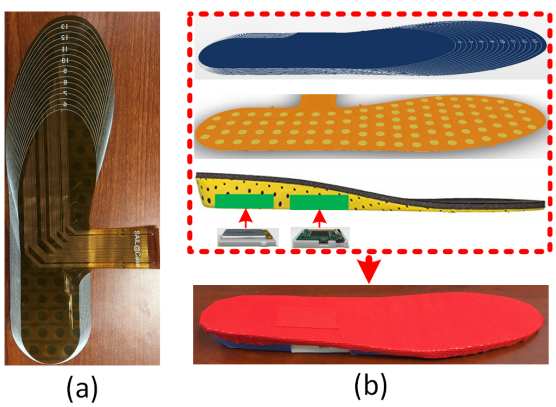

Fig. 2. The smart insole system used in this research. (a) An insole shaped pressure sensor array which has 96 pressure sensors uniformly distributed on it. (b) The assembly structure of the smart insole.

The smart insole was based on an insole-shaped flexible pressure sensor array which had 96 pressure sensors uniformly distributed on it. This pressure sensor array has a customizable design, which can be trimmed to fit different foot sizes [10]. Fig.2 (a) shows the pressure sensor array which can be trimmed to fit foot sizes from 5.5 US to 14 US. As shown in Fig. 2 (b), the smart insole has three layers: the fabric layer on the top, the sensor array in the middle, and the insole shaped case with a battery and a circuit board at the bottom. Through combining these three layers, the assembled smart insole can be used in the same way as a normal insole. Pressure sensor data was sampled at 30Hz, transmitted via Bluetooth, and finally collected by an Android application.

B. Optimal Pressure Sensor Locations Identification

To find out the optimal pressure sensor locations, it is important to understand how heel-strike and toe-off are detected. As shown in Fig. 3, the blue curve indicates normalized ground reaction force (GRF) with body weight (BW). GRF was calculated by adding the pressure of all the pressure sensors on the sensor array. During the stance phase when the foot is in contact with the ground, the measured GRF is high. During the swing phase when the foot is in the air, the measured GRF is low. Based on theses characteristics, thresholds were popularly used to identify heel-strike and toe-off. In this research, an adaptive threshold was used to increase the accuracy of heel-strike and toe-off detection [2]. As shown in Fig. 3, the dashed black line indicates the adaptive threshold. The first GRF sample (in a gait cycle) that is higher than the threshold will be recognized as heel-strike and the first GRF sample (in a gait cycle) that is lower than the threshold will be recognized as toe-off. Therefore, sample “B” and sample “D” indicates heel-strike and toe-off, respectively.

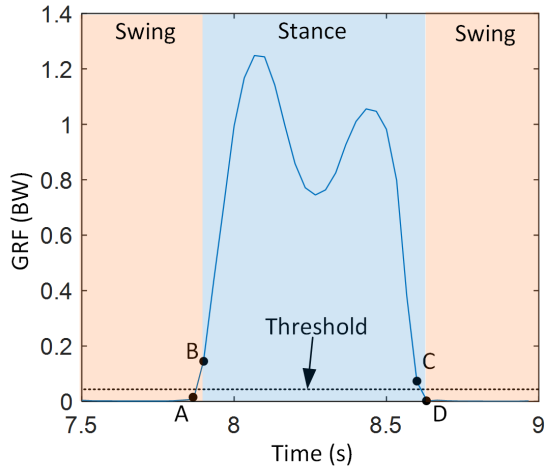

Fig. 3. Heel-strike and toe-off detection method. The blue line indicates the GRF. The area with blue background indicates the stance phase, and the area with orange background indicates the swing phase. The black dashed line indicates the threshold for heel-strike and toe-off detection. Points “A” and “B”, and points “C” and “D” are adjacent GRF samples, respectively.

Based on the mechanism of detecting heel-strike and toe-off, it is reasonable to deduce that (1) from sample “A” to sample “B”, sensor locations that have a larger pressure increase will have a larger contribution to the detection of heel-strike; (2) from sample “C” to sample “D”, sensor locations that have a larger pressure decrease will have a larger contribution for the detection of toe-off. Please note that sample “A” and sample “B”, and sample “C” and sample “D” are adjacent samples, respectively. Therefore, to find out optimal sensor locations for heel-strike and toe-off detection, sensor locations that have larger pressure changes between sample “A” and sample “B”, and between sample “C” and sample “D” will be selected.

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C. Data collection and processing

To avoid the potential errors introduced by one individual subject, six healthy subjects with normal gait patterns were involved in the experiment. The smart insole used in the experiment was trimmed to 10.5 US to fit the foot size of the subjects. As shown in Fig. 4, the red dashed line indicates the edge of the 10.5 US smart insole. All the sensors outside the edge (e.g. sensor 96) were trimmed off. Each sensor on the sensor array has its own identity (ID) number which is banded to one specific location. Through finding out the IDs of the optimal sensors, the optimal sensor locations could be identified. During the experiment, each subject wore the smart insole and walked on level ground straightly at their comfort speed for 15 strides. The heel-strike and toe-off detected with all the sensors on the sensor array were used as ground truth.

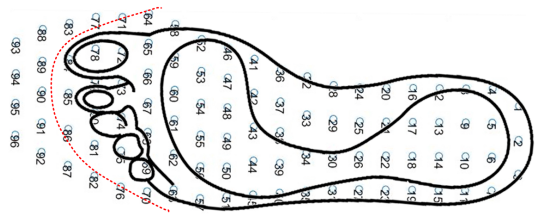

Fig. 4. Foot anatomical areas and sensor locations. The numbers are sensor ID. Small blue circles indicate the sensor locations. The red dashed curve indicates the edge of the 10.5 US smart insole.

The method introduced in the section II.B was used to find optimal sensor locations. To increase the reliability of the final results, for each stride, the top three optimal sensor locations for heel-strike and toe-off detection were extracted, respectively. To reveal the global optimal sensor locations that work best for all the subjects, extracted optimal sensor locations from 6 subjects for heel-strike and toe-off detection were combined together, respectively.

III. Results and Discussion

As shown in Fig. 5, sensor 14 was the global optimal sensor for heel-strike detection, because its occurrence frequency is the highest which accounted for 19.7% of the total occurrences of all the detected optimal sensors. In terms of the frequency, sensors 5, 6, 8, 10, 11, and 13 were also top-ranked optimal sensors which accounted for 7.2%, 7.7%, 5.7%, 11.4%, 9.9%, and 11.2%, respectively. Through considering both the occurrence frequency and sensor locations (Fig. 4) together, a trend could be revealed that sensors with a smaller distance to the global optimal sensor 14 had a higher occurrence frequency. These results indicate that the heel area is critical for heel-strike detection, and sensor locations that are closer to sensor 14 are better choices.

Similarly, as shown in Fig. 6, sensor 61 was the global optimal sensor for toe-off detection, which occurrence frequency accounted for 25.1% of the total occurrences of all the detected optimal sensors. In terms of the frequency, sensors 55, 59, 60 under metatarsal bones were top-ranked optimal sensors,

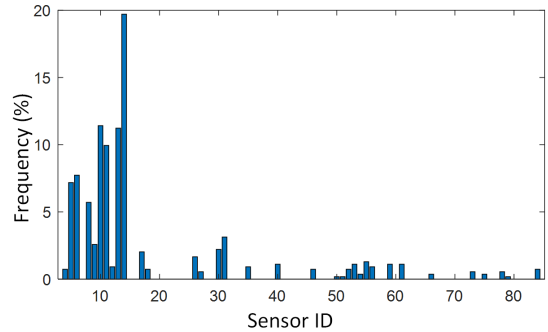

Fig. 5. Distribution and frequency of optimal sensors for heel-strike detection.

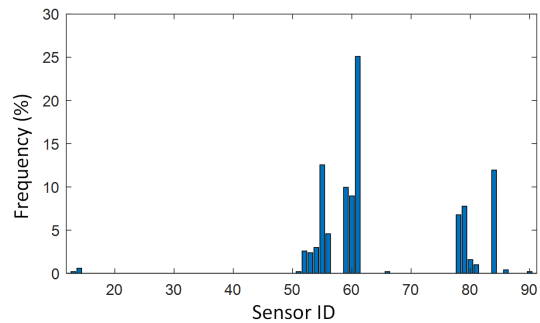

Fig. 6. Distribution and frequency of optimal sensors for toe-off detection.

which accounted for 12.5%, 10.0%, and 9.0%, respectively. In addition, sensors 78, 79, and 84 under the big toe area were also top-ranked optimal sensors, which accounted for 6.8%, 7.8%, and 12.0%, respectively. This result indicates that the front edge under metatarsal bones (indicated with sensors 55, 59, 60, and 61) and the area under the big toe (indicated with sensor 78, 79, and 84) could contribute to toe-off detection.

To further understand the contribution of different optimal sensors and three different anatomical areas in detecting heel-strike and toe-off, the performances of different sensor combinations were tested. As shown in Table I, the results of 7 tests were listed. The tested sensor combinations were: Test 1: sensors 14 and 61; Test 2: sensors 5, 6, 8, 10, 11, 13, 14, 55, 59, 60, 61; Test 3: sensors 5 and 61; Test 4: sensors 10 and 61; Test 5: sensors 14 and 55; Test 6: sensors 14 and 59; and Test 7: sensors 14 and 84. "Correct" indicates the percentage of detected Heel-strike or Toe-off that was the same as ground truth. "1 Sample Diff" indicates the percentage of detected Heel-strike or Toe-off that has one sample period difference with the ground truth. "1 Sample Diff" equaled a difference of 33ms because the sample rate of the pressure sensors was 30Hz. Since the accuracy of the ground truth for heel-strike and toe-off detection was 0.0 ± 14.1 ms and 5.2 ± 15.5 ms, respectively, one sample period difference with the ground truth did not mean the detected heel-strike or toe-off were wrong [2]. In addition, at the very beginning of heel-strike and toe-off, the pressure under foot is very close to the pressure of the swing phase. A slight change in the sensitivity

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of the dynamic threshold, which could be influenced by sensor numbers, will lead to this "1 Sample Diff". Considering these two reasons, this "1 Sample Diff" was not recognized as error in the following analysis about finding optimal sensor locations. The result analysis will use "Total", which is the summation of "Correct" and "1 Sample Diff", to evaluate the performance of different sensor locations. For each test, the results shown in the table were averaged results of all the six subjects.

TABLE I
THE ACCURACY OF HEEL-STRIKE AND TOE-OFF DETECTION WITH DIFFERENT SENSOR COMBINATIONS

Tests[a]	Heel-strike			Toe-off		
	Correct (%)	*1 Sample Diff(%)*	*Total (%)*	*Correct (%)*	*1 Sample Diff(%)*	*Total (%)*
Test 1	66.12	30.56	96.68	83.89	12.23	96.12
Test 2	90.01	7.23	97.24	95.01	3.34	98.35
Test 3	64.45	29.45	93.90	81.67	16.12	97.79
Test 4	74.45	20.56	95.01	86.67	11.12	97.79
Test 5	82.23	14.45	96.68	78.89	18.33	97.22
Test 6	75.00	22.23	97.23	72.67	23.34	96.01
Test 7	68.89	25.00	93.89	55.56	18.89	74.45

[a]Test 1: sensors 14 and 61; Test 2: sensors 5, 6, 8, 10, 11, 13, 14, 55, 59, 60; Test 3: sensors 5 and 61; Test 4: sensors 10 and 61; Test 5: sensors 14 and 55; Test 6: sensors 14 and 59; Test 7: sensors 14 and 84.

The performance of the global optimal sensors, which were sensors 14 and 61, were tested in "Test 1". The "Total" accuracy was above 96% for both heel-strike and toe-off, which indicated that the global optimal sensor locations were effective in detecting heel-strike and toe-off. In "Test 2", all the optimal sensors in the heel area (sensors 5, 6, 8, 10, 11, 13, and 14) and metatarsal bones area (sensors 55, 59, 60, and 61) were tested. Through comparing the results of "Test 1" and "Test 2", it is obvious that there was no significant change in the "Total" accuracy for both heel-strike and toe-off. This can be explained by the fact that each of the optimal pressure sensors were able to detect the pressure change during heel-strike or toe-off. Since two global optimal sensors were able to detect the pressure changes in time already, adding the other optimal pressure sensors would not increase the accuracy.

To further estimate the performance of the other top optimal sensors in the heel and metatarsal bones areas, four different sensor combinations were tested. Each sensor combination included one of the representative optimal sensors (i.e. sensors 5, 10, 55, and 59) and one of the global optimal sensors (i.e. sensors 14 and 61). From "Test 3" to "Test 6", the sensor combinations of sensors 5 and 61, sensors 10 and 61, sensors 14 and 55, and sensors 14 and 59 were tested, respectively. For heel-strike detection, the "Total" accuracy of "Test 3" and "Test 4" were 93.9% and 95.01%, respectively, which indicated that the other top optimal sensors (e.g. sensor 5 and sensor 10) in the heel area also had a good performance for heel-strike detection. Considering the differences in accuracy, this result also showed that sensor locations closer to sensor 14 had a better performance. This is in consistence with the conclusion at the beginning of this section. For toe-off detection, the "Total" accuracy of all the tests were higher than 96%. This result showed that the other top optimal sensors at the front edge of the metatarsal bones area also have a good performance for toe-off detection.

To test the performance of the big toe area in toe-off detection, the combination of sensors 14 and 84 was evaluated in "Test 7". As shown in Table I, the "Total" accuracy of toe-off detection was 74.45% which was much lower than the accuracy of the optimal sensors in the metatarsal bones areas. This result indicated that compared with the metatarsal bones area, the big toe area was not a good choice for toe-off detection during normal walking.

In summary, (1) two sensors are enough to have a good performance for heel-strike and toe-off detection; (2) for heel-strike detection, sensor locations in the heel area defined by sensors 5, 6, 8, 10, 11, 13, and 14 have good performance. For better performance, sensor locations closer to sensor 14 are better choices; (3) for toe-off detection, sensor locations at the front edge of the metatarsal bones area, which was indicated by sensors 55, 59, 60, and 61, have good performance. (4) It is not necessary to place sensors in the big toe area for toe-off detection during normal walking.

IV. CONCLUSION

In this research, optimal pressure sensor locations under the foot for heel-strike and toe-off detection were explored. The contributions of different anatomy foot areas were discussed. The results of this research could be important references for future researcher to effectively use pressure sensors for smart insole design and gait analysis.

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